LEA DEMO / 01

A tiny mathematical project

1 Doubling natural numbers

We start with a definition, prove a structural fact, and use it in a corollary. The mathematics is deliberately small so we can focus on the formalization workflow.

Definition 1.1 (Doubling)**.** *For a natural number $n \in \mathbb{N}$, define $D(n) = n + n$.*

Lemma 1.2 (Doubling zero)**.** *The double of zero is zero: $D(0) = 0$.*

Proof. By the definition of D, we have $D(0) = 0 + 0 = 0$. □

Claim 1.3 (Doubling a sum)**.** *For every $m, n \in \mathbb{N}$,*

$$D(m + n) = D(m) + D(n).$$

Proof. Expand the definition and rearrange addition:

$$(m + n) + (m + n) = (m + m) + (n + n).$$

□

Corollary 1.4 (Adding two before doubling)**.** *For every $n \in \mathbb{N}$,*

$$D(n + 2) = D(n) + 4.$$

Proof. Apply the preceding claim with $m = n$ and with its second argument equal to 2. Then use $D(2) = 2 + 2 = 4$. □

Definition ⟶ structural fact ⟶ corollary. The source document remains ordinary LaTeX.

When the source needs attention

2 A deliberately flawed draft

This exercise starts with an intentionally false claim and an invalid proof. The presenter notes give the replacement text to use after inspecting Lea's finding.

Theorem 2.1 (Square versus the original number)**.** *For every real number $x \in \mathbb{R}$,*

$$x^2 \geq x.$$

Proof. We have $x^2 - x = x(x-1)$. Both factors are nonnegative for every real number x, so their product is nonnegative. Therefore $x^2 \geq x$. □

The aim is to inspect the reported issue, clarify the mathematical statement, and bring the formal artifact back into agreement with the revised source.

Read the statement, the author's proof, the Lean artifact, and the scope of the assessment together.

One supporting result, two consumers

3 A shifted square

Definition 3.1 (Shifted square)**.** *For $a, x \in \mathbb{R}$, set $Q_a(x) = (x - a)^2$.*

Lemma 3.2 (Shift identity)**.** *For all $a, t \in \mathbb{R}$,*

$$Q_a(a + t) = t^2.$$

Proof. Substitute the definition and simplify $(a + t) - a = t$. □

Theorem 3.3 (One unit from the center)**.** *For every $a \in \mathbb{R}$, $Q_a(a + 1) = 1$.*

Proof. Apply the Shift identity with $t = 1$, then simplify $1^2 = 1$. □

Theorem 3.4 (Two units from the center)**.** *For every $a \in \mathbb{R}$, $Q_a(a + 2) = 4$.*

Proof. Apply the same Shift identity with $t = 2$, then simplify $2^2 = 4$. □

The unmarked identity supplies source context. The shared-results view shows whether a formal helper was actually created and reused.

Two optional input forms

4 Standalone prose and existing Lean

The main demo uses comments and tags inside environments. These two small extras show that an enclosing theorem environment is optional and that existing Lean can also be supplied as the source material.

4.1 A standalone proposition tag

The double of one is two: $D(1) = 2$.

4.2 A Lean code block

The following is a statement with a complete proof. The code is the input to Lea; its presence in the PDF alone is not evidence of a checked Lea run.

```
theorem demo_nat_zero (n : Nat) : n + 0 = n := by
  rfl
```

These examples are optional in a 15-minute recording. They can also supply small independent items for batch work.